

EXPERIMENT NO: - 03

AIM OF THE EXPERIMENT: - TO STUDY AND MEASURE

- (I) THE FIELD RESISTANCE AND
- (II) ARMATURE RESISTENCE OF A DC SHUNT MOTOR.

(I) MEASUREMENT OF SHUNT FIELD RESISTENCE (HIGH VALUE)

THEORY: -

DC motor is an Electrical device, which convert DC electrical energy to mechanical energy. It works under the principle that, whenever a current carrying conductor is placed in a magnetic field it experiences a mechanical force. The essential parts of a DC machine are (i) Yoke, (ii) Field system, (iii) Armature core, (iv) Armature winding, (v) Commutator, and (vi) Brushes.

The field system consist of three parts (i) Pole core, (ii) pole shoe and (iii) magnetizing coil or field coil.

MAGNETIZING COIL: -

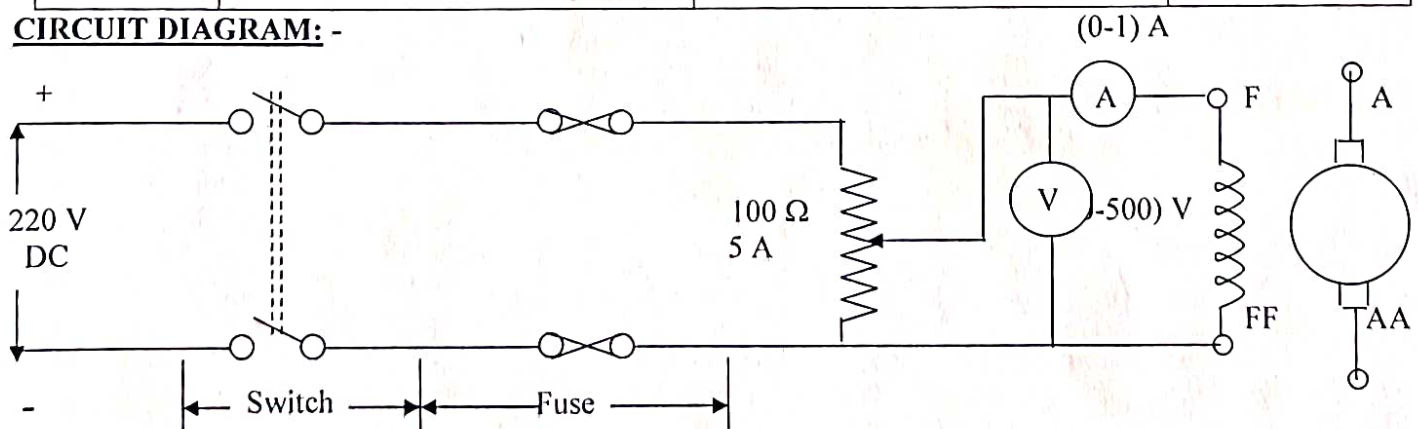
Enamelled copper wire is used for the construction of field coils. Coils are wound on a wooden former and then placed around the pole core. Field coils are of two types (i) shunt field coil and (ii) series field coil.

Shunt coil is of high resistance having many turns of fine wire. Where as series coil is of small resistance having few turns of wire of large cross-sectional area.

APPARATUS REQUIRED: -

SL. NO.	NAME OF THE APPARATUS	SPECIFICATION	QUANTITY
1	DC MOTOR	2 KW, 220 V, 9 A	01
2	VOLTMETER	(0-500) V, PMMC TYPE	01
3	AMMETER	(0-1) A, PMMC TYPE	01
4	RHEOSTAT	(0-100) Ω , 5 A	01
5	CONNECTING WIRES	3/22 SWG, INSULATED CU WIRE	AS REQUIRED

CIRCUIT DIAGRAM: -



PROCEDURE: -

1. MAKE THE CONNECTION AS PER THE CIRCUIT DIAGRAM.
2. SWITCH ON THE SUPPLY AND GRADUALLY INCREASE THE VOLTAGE IN SUITABLE STEPS.
3. TAKE THE CORRESPONDING READINGS OF VOLTMETER AND AMMETER.
4. REPEAT STEPS 2-3 AND TAKE 5-6 READINGS.

TABULATION: -

SL. NO	AMMETER READING (I) IN AMP	VOLTMETER READING (V) IN VOLT	FIELD RESISTANCE $R_{SH} = V/I$ IN Ω	MEAN R_{SH} IN Ω N $R_{SH} = (\sum R_{SHi})/N$ $i=1$
1				
2				
3				
4				
5				
6				

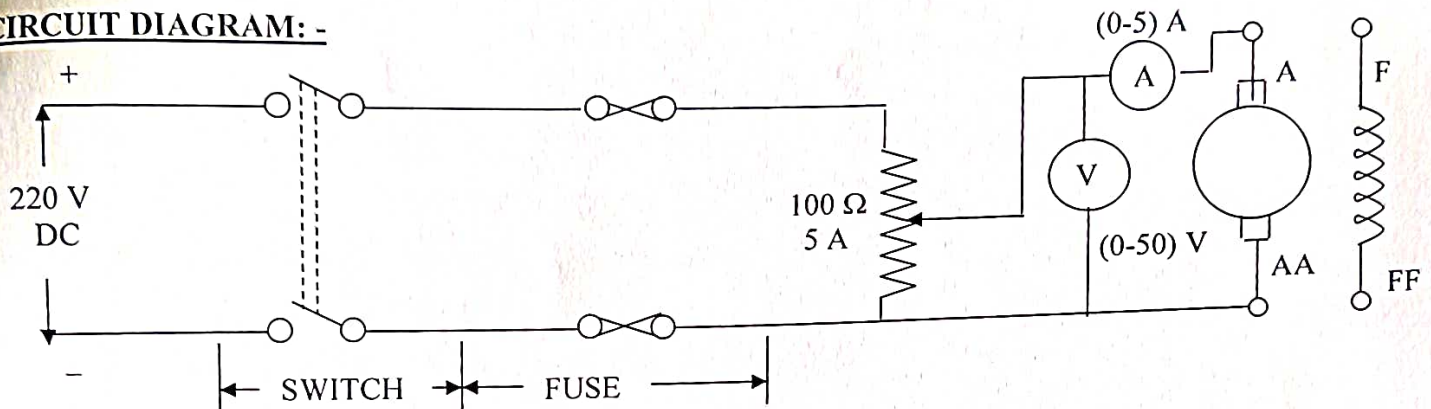
(I) MEASUREMENT OF ARMATURE RESISTANCE (LOW VALUE)**THEORY: -****ARMATURE WINDING: -**

It is the heart of the DC machine. It is placed where conversion of power takes place. It is an arrangement of conductors to develop the desired emf by relative motion in a hetero-polar magnetic field. In armature winding conductor or group of conductors are distributed in different way in slots all over the periphery of armature. The conductor may be connected in series and parallel depending upon the current and voltage requirement. There are two types of armature winding, (i) wave winding, (ii) lap winding. In wave winding conductors are divided in two parallel paths irrespective of the number of pole of the machine. In lap winding the conductors are connected in such a way that the number of parallel paths are equal to number of poles.

APPARATUS REQUIRED: -

SL. NO.	NAME OF THE APPARATUS	SPECIFICATION	QUANTITY
1	DC MOTOR	2 KW, 220 V, 9 A	01
2	VOLTMETER	(0-50) V, PMMC TYPE	01
3	AMMETER	(0-5) A, PMMC TYPE	01
4	RHEOSTAT	(0-100) Ω , 5 A	01
5	CONNECTING WIRES	3/22 SWG, INSULATED CU WIRE	AS REQUIRED

CIRCUIT DIAGRAM: -



PROCEDURE: -

1. MAKE THE CONNECTION AS PER THE CIRCUIT DIAGRAM.
2. SWITCH ON THE SUPPLY AND GRADUALLY INCREASE THE VOLTAGE IN SUITABLE STEPS.
3. TAKE THE CORRESPONDING READINGS OF VOLTMETER AND AMMETER.
4. REPEAT STEPS 2-3 AND TAKE 5-6 READINGS.

TABULATION: -

SL. NO	AMMETER READING (I) IN AMP	VOLTMETER READING (V) IN VOLT	ARMATURE RESISTANCE $R_{SE} = V/I$ IN Ω	MEAN R_a
				IN Ω N $R_a = (\sum R_{ai})/N$ $i=1$
1				
2				
3				
4				
5				
6				

RESULTS: -

CONCLUSION: - (i) Average value of field resistance = $R_{sh} =$ _____
(ii) Average value of armature resistance = $R_a =$ _____.

PRECAUTIONS: -

1. ALL THE CONNECTION SHOULD BE RIGHT & TIGHT.
2. THE VOLTMETER & AMMETER SHOULD BE CAREFULLY CHOSEN SO THAT THEIR RANGE ARE MORE THAN THE MAXIMUM VALUE TO BE MEASURED.
3. THE FIELD CURRENT SHOULD NOT BE INCREASED TO EXCEED THE RATED VALUE.
4. SUPPLY SHOULD BE SWITCHED-ON AFTER ENSURING CORRECTNESS OF CONNECTIONS.
5. ANY LIVE TERMINAL SHOULD NOT BE TOUCHED WHILE SUPPLY IS ON.

QUESTIONS: -

1. What are the types of DC motor?
2. What is the function of Commutator in DC motor?
3. What is back EMF and what is its significance?
4. How can you reverse the direction of rotation of a DC shunt motor?
5. Why the armature resistance of a DC shunt motor is less than the field resistance?]
6. What is the order of armature resistance and field resistance in (i) shunt machine and (ii) series machine?